thinking in terms of the ‘good of the species’ view of evolution that we often forget to ask perfectly reasonable questions like: Why don’t lions hunt other lions? Another good question of a type which is seldom asked is: Why do antelopes run away from lions instead of hitting back?

The reason lions do not hunt lions is that it would not be an ESS for them to do so. A cannibal strategy would be unstable for the same reason as the hawk strategy in the earlier example. There is too much danger of retaliation. This is less likely to be true in contests between members of different species, which is why so many prey animals run away instead of retaliating. It probably stems originally from the fact that in an interaction between two animals of different species there is a built-in asymmetry which is greater than that between members of the same species. Whenever there is strong asymmetry in a contest, ESSs are likely to be conditional strategies dependent on the asymmetry. Strategies analogous to ‘if smaller, run away; if larger, attack’ are very likely to evolve in contests between members of different species because there are so many available asymmetries. Lions and antelopes have reached a kind of stability by evolutionary divergence, which has accentuated the original asymmetry of the contest in an ever-increasing fashion. They have become highly proficient in the arts of, respectively, chasing, and running away. A mutant antelope that adopted a ‘stand and fight’ strategy against lions would be less successful than rival antelopes disappearing over the horizon.

I have a hunch that we may come to look back on the invention of the ESS concept as one of the most important advances in evolutionary theory since Darwin.* It is applicable wherever we find conflict of interest, and that means almost everywhere. Students of animal behaviour have got into the habit of talking about something called ‘social organization’. Too often the social organization of a species is treated as an entity in its own right, with its own biological ‘advantage’. An example I have already given is that of the ‘dominance hierarchy’. I believe it is possible to discern hidden group selectionist assumptions lying behind a large number of the statements that biologists make about social organization. Maynard Smith’s concept of the ESS will enable us, for the first time, to see clearly how a collection of independent selfish entities can come to resemble a single organized whole. I think this will be true not only of social organizations within species, but also of